

Submodule preservers on Hilbert modules with two orthonormal vectors

Partial-result packet for `fa_banach_001`
`agent_lane_03`, model GPT5.6

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Abstract

Sharifi asks when an adjointable operator on a Hilbert C^* -module that leaves every closed submodule invariant must be a coefficient morphism, and poses a compact-operator variant. We give an exact answer for every right Hilbert module containing two orthonormal module vectors: the preservers are precisely right multiplication by central coefficients. No injectivity, surjectivity, or nonzero assumption is required. Consequently the compact preservers on $\mathcal{A}^n$, $n \geq 2$, are exactly the central scalar matrices, while the only compact preserver on the standard module $\ell^2(\mathcal{A})$ is zero. This is a proved partial answer; arbitrary Hilbert modules without such a pair remain outside its scope.

1 Source questions and their literal boundary cases

Sharifi's arXiv:2506.01161 asks in Problem 2.16 (PDF page 11) for which C^* -algebras universal closed-submodule invariance forces an adjointable operator to equal $I.a$ for a *nonzero* coefficient a [1].

INVARIANT SUBMODULES OF MODULAR OPERATORS

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We know that if every subspace of a Hilbert space is invariant under a linear transformation T , then T is a scalar transformation. This fact motivates us to close this section with the following problem. We will study such C^* -algebras and Hilbert C^* -modules elsewhere.

Problem 2.16. *Let E be a Hilbert $\mathcal{A}$ - $\mathcal{A}$ -bimodule. For which C^* -algebras $\mathcal{A}$ does the following assertion hold?*

If every closed submodule of E is invariant under a bounded adjointable operator $T : E \rightarrow E$, then T is a C^ -coefficient morphism, i.e., there exists a nonzero element $a \in \mathcal{A}$ such that $T = I.a$, where I is the identity operator on E .*

3. C^* -ALGEBRAS OF COMPACT OPERATORS

Suppose that $\mathcal{A}$ is an arbitrary C^* -algebra of compact operators, i.e. $\mathcal{A}$ is a C^* -subalgebra

Literally, $T = 0$ satisfies the hypothesis, but on the faithful standard module $\mathcal{A}^2$ it cannot equal right multiplication by a nonzero coefficient. The natural repair is to allow $a = 0$, or to assume $T \neq 0$.

Problem 4.10 (PDF page 17) asks for the compact analogue under the condition $I \neq K$.

Corollary 4.9. *Let $\mathcal{A}$ be a finite dimensional C^* -algebra and let E be a Hilbert $\mathcal{A}$ -module that is not finitely generated. If $S \in \mathcal{L}(E)$ commutes with a nonzero compact operator K on E , then S has a proper nonzero invariant submodule. In particular, every nonzero compact operator on a Hilbert $\mathcal{A}$ -module that is not finitely generated, has a proper nonzero invariant submodule.*

We close our paper with the following problem which is under study in some special cases.

Problem 4.10. *Let E is a Hilbert $\mathcal{A}$ -module and let $I \neq K \in \mathcal{K}_{\mathcal{A}}(E)$. Characterize those C^* -algebras $\mathcal{A}$ for which every closed submodule of E is invariant under K .*

That condition still permits $K = 0$, which automatically preserves every submodule. Below we classify the natural repaired problem on standard and, more generally, two-orthonormal-vector modules.

2 Idea of the Proof

The Hilbert-space proof tests one-dimensional spans. Here the replacement is a family of closed cyclic graph submodules. The coordinate submodules $e_1\mathcal{A}, e_2\mathcal{A}$ first diagonalize the operator. The diagonal graph $(e_1 + e_2)\mathcal{A}$ equates the two diagonal coefficients. The graphs $(e_1 + e_2c)\mathcal{A}$, for arbitrary $c \in \mathcal{A}$, force that common coefficient to commute with all of $\mathcal{A}$. Finally, $(e_1 + y)\mathcal{A}$ propagates the same coefficient to the orthogonal complement. Unlike the known general theorem for bijective preservers, this argument uses no injectivity at all.

3 Exact preserver theorem

Let $\mathcal{A}$ be a unital C^* -algebra and E a right Hilbert $\mathcal{A}$ -module. We use the convention that the inner product is linear in the second variable. For $z \in Z(\mathcal{A})$, let $R_z(x) = xz$.

Theorem 1. *Suppose E contains vectors e_1, e_2 with*

$$\langle e_i, e_j \rangle = \delta_{ij} 1_{\mathcal{A}}.$$

For $T \in \mathcal{L}(E)$, the following are equivalent:

1. *every closed right $\mathcal{A}$ -submodule of E is T -invariant;*
2. *$T = R_z$ for a unique $z \in Z(\mathcal{A})$.*

Proof. Assume (1). The coordinate submodules $e_i\mathcal{A}$ are closed, so

$$Te_1 = e_1b_1, \quad Te_2 = e_2b_2$$

for unique $b_1, b_2 \in \mathcal{A}$. The cyclic submodule $(e_1 + e_2)\mathcal{A}$ is closed and invariant. Hence $e_1b_1 + e_2b_2 = (e_1 + e_2)b$ for some $b \in \mathcal{A}$; taking inner products with e_1, e_2 gives $b_1 = b_2 = b$.

For any $c \in \mathcal{A}$, the map

$$a \longmapsto (e_1 + e_2c)a$$

is bounded below, since

$$\langle e_1 + e_2c, e_1 + e_2c \rangle = 1 + c^*c \geq 1.$$

Its range $W_c = (e_1 + e_2c)\mathcal{A}$ is therefore closed. Invariance gives

$$e_1b + e_2bc = T(e_1 + e_2c) = (e_1 + e_2c)a_c.$$

The first coordinate yields $a_c = b$, and the second yields $bc = cb$. As c was arbitrary, $b \in Z(\mathcal{A})$.

Let $M = e_1\mathcal{A} \oplus e_2\mathcal{A}$ and $F = M^\perp$. The finite orthonormal summand M is complemented, so $E = M \oplus F$. Since F is a closed submodule, $T(F) \subseteq F$. For $y \in F$, the same lower-bound argument shows that $W_y = (e_1 + y)\mathcal{A}$ is closed. Thus

$$e_1b + Ty = T(e_1 + y) = (e_1 + y)a_y.$$

Uniqueness in $e_1\mathcal{A} \oplus F$ gives $a_y = b$ and $Ty = yb$. Right $\mathcal{A}$ -linearity and centrality of b now show $Tx = xb$ for every $x \in E$. Thus $T = R_b$.

Conversely, if $z \in Z(\mathcal{A})$, then R_z is adjointable and right $\mathcal{A}$ -linear, with adjoint R_{z^*} . Every right submodule is closed under right multiplication by z , so it is R_z -invariant. Uniqueness follows from $e_1z = R_z(e_1)$. $\square$

4 Finite-free and standard-module consequences

Corollary 1. *Let $\mathcal{A}$ be a nonzero unital C^* -algebra.*

1. *For $2 \leq n < \infty$, the operators in $\mathcal{L}(\mathcal{A}^n) = M_n(\mathcal{A})$ leaving every closed submodule invariant are exactly zI_n with $z \in Z(\mathcal{A})$. Since $\mathcal{K}_{\mathcal{A}}(\mathcal{A}^n) = M_n(\mathcal{A})$, this is also the exact compact classification.*
2. *On the standard module $H_{\mathcal{A}} = \ell^2(\mathcal{A})$, the only $K \in \mathcal{K}_{\mathcal{A}}(H_{\mathcal{A}})$ leaving every closed submodule invariant is $K = 0$.*
3. *The universal-over-all- E, K reading of Problem 4.10 fails for every such $\mathcal{A}$: on $E = \mathcal{A}^2$, the compact projection $K = \text{diag}(1, 0) \neq I$ does not preserve $(e_1 + e_2)\mathcal{A}$.*

Proof. The first assertion is Theorem 1. For the second, that theorem gives $K = R_z$ with $z \in Z(\mathcal{A})$. If (e_m) is the standard orthonormal sequence, then $\|R_z e_m\| = \|z\|$ for every m . On the other hand, $\|F e_m\| \rightarrow 0$ for every $F \in \mathcal{K}_{\mathcal{A}}(H_{\mathcal{A}})$: this is immediate for finite sums of rank-one operators because the coordinates of every vector in $\ell^2(\mathcal{A})$ tend to zero, and follows for compact operators by norm approximation. Therefore $z = 0$. Finally, $K(e_1 + e_2) = e_1 \notin (e_1 + e_2)\mathcal{A}$, proving (3). $\square$

5 Relation to the later partial answer

Frank's arXiv:2507.11206 explicitly cites Sharifi's Problem 2.16 and proves that on an arbitrary full Hilbert module every *bijective* bounded module morphism preserving all norm-closed submodules is multiplication by an invertible central multiplier [2, Thm. 1.2]. Frank also states that a complete result for merely injective preservers is out of reach. Theorem 1 is complementary: it permits arbitrary noninjective operators, but assumes that the module contains two orthonormal vectors. General modules without such a pair remain open here.

Verification, novelty scope, and review recommendation

The exact script `code/check_matrix_models.py` verifies for $M_d(\mathbb{Q})$, $1 \leq d \leq 7$, that commutation with all matrix units has one-dimensional solution space, and checks the diagonal-projection witness. The proof itself has no computational dependency.

The four local run indexes and current arXiv searches were checked using both arXiv ids, the exact problem text, “ C^* -submodule preserving”, “Alg Lat”, central multipliers, orthonormal module vectors,

and non-bijective variants. They found Frank's bijective theorem but no statement matching Theorem 1 or Corollary 1. This is bounded novelty evidence, not a publication-novelty guarantee. Human review should focus on closedness of the graph submodules, the orthogonal-complement step, and the compactness argument on $\ell^2(\mathcal{A})$.

References

- [1] K. Sharifi, *Invariant submodules of modular operators and Lomonosov type theorem for Hilbert C^* -modules*, arXiv:2506.01161, <https://arxiv.org/abs/2506.01161>.
- [2] M. Frank, *C^* -submodule preserving module mappings on Hilbert C^* -modules*, arXiv:2507.11206, <https://arxiv.org/abs/2507.11206>.